Lab sheet-3

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import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from scipy import stats

import statsmodels.api as sm

# Step 1 & 2: Load the Dataset and Import Libraries

filename = "HR Employee-Attrition.csv"

df = pd.read_csv(filename)

print(f"Dataset successfully loaded. Shape: {df.shape}\n")

# Step 3: Descriptive Statistics for Numerical Features

numerical_cols = ['Age', 'MonthlyIncome', 'TotalWorkingYears', 'YearsAtCompany'] #

Mean, Median, Variance, Standard Deviation

desc_stats = df[numerical_cols].agg(['mean', 'median', 'var', 'std']).T

# Compute Mode

desc_stats['mode'] = [df[col].mode()[0] for col in numerical_cols]

print("=" * 60)

print("DESCRIPTIVE STATISTICS")

print("=" * 60)

print(desc_stats.round(2))

print()

# Step 4: Pearson Correlation Analysis & Automatic Plot Saving

corr_matrix = df[numerical_cols].corr(method='pearson')

print("=" * 60)

print("PEARSON CORRELATION MATRIX")

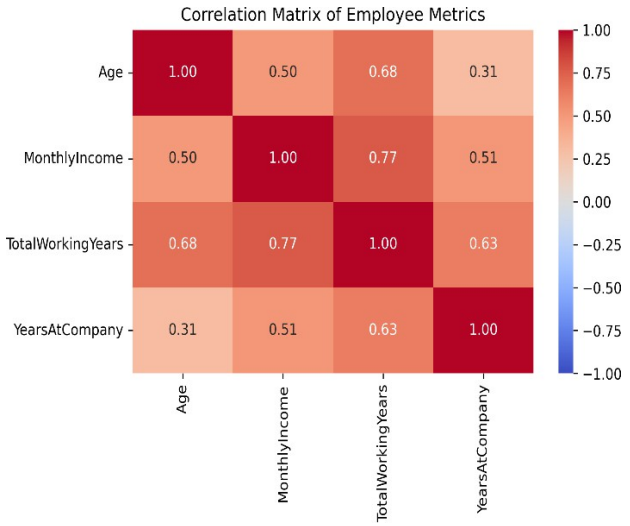
print("=" * 60)

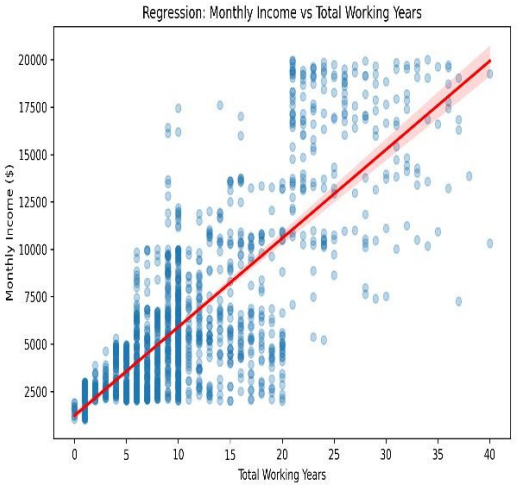
print(corr_matrix.round(3))
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print()
# Generate and auto-save Heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", vmin=-1, vmax=1)
plt.title("Correlation Matrix of Employee Metrics")
plt.tight_layout()
# Automatically save before showing
plt.savefig("correlation_matrix.png", dpi=300, bbox_inches='tight')
print("[INFO] Figure saved automatically as 'correlation_matrix.png'")
# Non-blocking show prevents terminal execution from halting
plt.show(block=False)
# Step 5 & 6: Independent Two-Sample t-Test
# Business Question: Does monthly income differ between employees who stay vs leave?
# H0: Mean MonthlyIncome(Attrition == 'Yes') == Mean MonthlyIncome(Attrition == 'No')
# H1: Mean MonthlyIncome(Attrition == 'Yes') != Mean MonthlyIncome(Attrition == 'No')
income_left = df[df['Attrition'] == 'Yes']['MonthlyIncome']
income_stayed = df[df['Attrition'] == 'No']['MonthlyIncome']
# Welch's t-test (equal_var=False)
t_stat, p_val_ttest = stats.ttest_ind(income_left, income_stayed, equal_var=False)
print("\n" + "=" * 60)
print("HYPOTHESIS TEST 1: TWO-SAMPLE T-TEST")
print("=" * 60)
print(f"Mean Income (Left):   ${income_left.mean():.2f}")
print(f"Mean Income (Stayed): ${income_stayed.mean():.2f}")
print(f"t-statistic:          {t_stat:.4f}")
print(f"p-value:              {p_val_ttest:.4e}")
if p_val_ttest < 0.05:
    print("Decision: Reject the Null Hypothesis (H0).")
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    print("Conclusion: Significant difference in monthly income between employees who leave and stay.")
else:
    print("Decision: Fail to reject the Null Hypothesis (H0).")
print()
# Step 7: One-Way ANOVA
# Business Question: Does monthly income vary across different Job Roles?
# H0: Mean MonthlyIncome is equal across all Job Roles
# H1: At least one Job Role has a different mean MonthlyIncome
role_groups = [group['MonthlyIncome'].values for _, group in df.groupby('JobRole')] f_stat,
p_val_anova = stats.f_oneway(*role_groups)
print("=" * 60)
print("HYPOTHESIS TEST 2: ONE-WAY ANOVA")
print("=" * 60)
print(f"F-statistic: {f_stat:.4f}")
print(f"p-value:     {p_val_anova:.4e}")
if p_val_anova < 0.05:
    print("Decision: Reject the Null Hypothesis (H0).")
    print("Conclusion: Monthly income differs significantly across job roles.")
else:
    print("Decision: Fail to reject the Null Hypothesis (H0).")
print()
# Step 8 & 9: Simple Linear Regression (OLS)
# Dependent (y): MonthlyIncome, Independent (X): TotalWorkingYears X
= sm.add_constant(df['TotalWorkingYears'])
y = df['MonthlyIncome']
model = sm.OLS(y, X).fit()
```

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print("=" * 60)
print("LINEAR REGRESSION ANALYSIS")
print("=" * 60)
print(model.summary())
# Plot and auto-save Regression Line
plt.figure(figsize=(8, 5))
sns.regplot(
    x='TotalWorkingYears',
    y='MonthlyIncome',
    data=df,
    scatter_kws={'alpha': 0.3},
    line_kws={'color': 'red'}
)
plt.title("Regression: Monthly Income vs Total Working Years")
plt.xlabel("Total Working Years")
plt.ylabel("Monthly Income ($)")
plt.tight_layout()
plt.savefig("regression_line.png", dpi=300, bbox_inches='tight')
print("[INFO] Figure saved automatically as 'regression_line.png'
plt.show()
```






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============================================================
DESCRIPTIVE STATISTICS
============================================================
                      mean  median          var      std  mode
Age                  36.92    36.0        83.46     9.14    35
MonthlyIncome      6502.93  4919.0  22164857.07  4707.96  2342
TotalWorkingYears    11.28    10.0        60.54     7.78    10
YearsAtCompany        7.01     5.0        37.53     6.13     5

============================================================
PEARSON CORRELATION MATRIX
============================================================
                     Age  MonthlyIncome  TotalWorkingYears  YearsAtCompany
Age                1.000          0.498              0.680           0.311
MonthlyIncome      0.498          1.000              0.773           0.514
TotalWorkingYears  0.680          0.773              1.000           0.628
YearsAtCompany     0.311          0.514              0.628           1.000

[INFO] Figure saved automatically as 'correlation_matrix.png'

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HYPOTHESIS TEST 1: TWO-SAMPLE T-TEST
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Mean Income (Left):   $4787.09
Mean Income (Stayed): $6832.74
t-statistic:          -7.4826
p-value:              4.4336e-13
Decision: Reject the Null Hypothesis (H0).
Conclusion: Significant difference in monthly income between employees who leave and stay
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HYPOTHESIS TEST 2: ONE-WAY ANOVA
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F-statistic: 810.2141
p-value:     0.0000e+00
Decision: Reject the Null Hypothesis (H0).
Conclusion: Monthly income differs significantly across job roles.

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LINEAR REGRESSION ANALYSIS
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                            OLS Regression Results
==============================================================================
Dep. Variable:          MonthlyIncome   R-squared:                       0.597
Model:                          OLS   Adj. R-squared:                  0.597
Method:               Least Squares   F-statistic:                     2178.
Date:              Sun, 06 Sep 2026   Prob (F-statistic):          2.73e-292
Time:                      13:11:09   Log-Likelihood:                -13848.
No. Observations:              1470   AIC:                         2.770e+04
Df Residuals:                  1468   BIC:                         2.771e+04
Df Model:                         1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1227.9353    137.299      8.944      0.000     958.612    1497.259
TotalWorkingYears   467.6584     10.021     46.669      0.000     448.002     487.315
==============================================================================
Omnibus:                       47.473   Durbin-Watson:                   1.993
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               79.304
Skew:                           0.269   Prob(JB):                     6.02e-18
Kurtosis:                       4.003   Cond. No.                         24.2
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